Brahmani Thota

Senior Data Engineer

Windsor Mill, MD | 908-290-7971 | LinkedIn | Email | Portfolio

SUMMARY

Senior Data Engineer with **5+ years** of experience designing, migrating, and optimising large-scale healthcare and financial data platforms. Proven expertise in **Snowflake, AWS, SQL, PySpark,** and **BI solutions**, delivering measurable business impact through automation, data quality improvements, and accelerated reporting cycles.

SKILLS

Programming & Querying: Python, SQL, PySpark, Scala, Java, Shell Scripting
Data Engineering & Processing: Apache Spark, Apache Airflow, dbt, ETL/ELT Pipelines, Data Modelling (Star & Snowflake Schema), Batch & Incremental Processing
Cloud & Platforms: AWS (S3, Glue, Athena, RDS, Lambda, EC2, IAM, DynamoDB), Snowflake, Palantir Foundry, Teradata
Databases & Storage: Relational Databases, Data Warehouses, Data Lakes
Orchestration & DevOps: CI/CD Pipelines, GitHub, Infrastructure as Code (Basics), DevOps Automation
Analytics & Visualisation: Power BI, Tableau, AWS QuickSight, Advanced Excel
Monitoring & Optimisation: Query Performance Tuning, Cost Optimisation, Pipeline Monitoring, Error Handling

EXPERIENCE

Centene Corporation

Data Engineer II – Production Team (Full Time) *Mar 2025 - Present*

- Led end-to-end **troubleshooting** and **optimization** of monthly financial reporting pipelines using **SQL, Snowflake, and dbt,** enforcing data quality issues early to reduce downstream defects and deployed reports to production 50% faster.
- Designed medical data validation frameworks using SQL and healthcare business rules.
- Built and maintained dbt models and tests to enforce data quality checks across Medicaid/Medicare financial datasets, reducing manual validation and improving trust in downstream reporting.
- Provisioned and deployed AWS Glue and S3 resources using **Terraform** to support Snowflake ingestion pipelines, streamlining infrastructure setup for Medicaid/Marketplace data workflows.
- Integrated AWS S3 and AWS Glue ingestion pipelines to catalog and preprocess Medicaid and Marketplace source data, improving financial accuracy, enabling metadata-driven validation, and reducing manual validation effort by 60%.
- Developed scalable **Power BI dashboards** identifying revenue leakage, duplicate claims, and expense mismatches, enabling finance stakeholders to resolve discrepancies earlier and accelerating month-end close cycles by approximately 40 per cent.
- Engineered consolidated financial reporting workflows across **Medicare** and **Marketplace** using **Snowflake** stored procedures, reducing report generation timelines from multiple days to under twenty-four hours consistently.
- Partnered with finance stakeholders in an **Agile delivery environment** to translate regulatory requirements into auditable **data logic**, proactively communicating across **cross-functional teams** while supporting production readiness reviews through documentation of pipeline dependencies and validation metrics, reducing recurring financial data incidents by 20%.

Data Engineer II – Snowflake Transition Team (Full Time) *Jan 2024 – Feb 2025*

- Led large-scale ETL/ELT migration from Palantir to Snowflake, redesigning data models and rewriting dbt, PySpark, and SQL pipelines to align with enterprise financial reporting standards.
- Automated financial report generation using **AWS Glue, Snowflake** stored procedures, and **DevOps scripting**, reducing reporting turnaround time by approximately 60 per cent across three healthcare lines of business.
- Consolidated **Medicare** and **Medicaid pipelines** by converting legacy **SQL datasets** into optimised **Python** and **PySpark workflows**, reducing dataset sprawl by 25 per cent and improving long-term maintainability.
- Conducted detailed data analysis and table-level mapping during migration activities, proactively identifying data gaps and ensuring parity between legacy platforms and **Snowflake-based** financial reports.
- Built **Tableau dashboards** on curated financial datasets to support trend analysis and executive reporting, improving self-service analytics adoption and reducing ad-hoc reporting requests from finance stakeholders.
- Collaborated with cross-functional engineering teams to implement **query performance tuning** strategies, improving Snowflake efficiency while lowering overall compute and storage costs.
- Worked with **Quiver** dashboards in **Palantir** to validate the financial reports of Medicare, Medicaid and Marketplace plans and debugged them using **Contour**.

Data Engineer I – Palantir Team (Contract) *Aug 2021 – Dec 2023*

- Engineered and validated high-volume healthcare financial datasets ranging from 120 to 150 gigabytes monthly using **Palantir Foundry,** supporting accurate Medicare, Medicaid, and Marketplace reporting.

- Resolved recurring data quality and pipeline failures using **Contour debugging** and **SQL analysis,** improving reporting reliability and reducing issue resolution cycles to under one week.
- Designed automated monthly **validation processes** for regulated healthcare plans, strengthening compliance requirements and improving stakeholder confidence in published financial and operational results.
- Built high-performance data extraction pipelines from **Teradata** to **REST APIs** using **Python** multiprocessing, reducing computation time by approximately 80 per cent and enabling faster downstream analytics.
- Partnered with finance analysts to translate complex reporting logic into **scalable data pipelines**, minimising manual interventions during peak reporting and audit periods.

IBM
Python Developer Intern - Quantum Computing (Full Time) *June 2020 to May 2021*

- Developed interactive **Python-based** learning platforms using **Anvil** and **Qiskit,** enabling hands-on experimentation with quantum gates, optimisers, and quantum machine learning concepts for global learners.
- Implemented hybrid classical and quantum optimisation algorithms for real-world problems, creating reusable **Python modules** adopted as educational references within IBM Quantum programs.
- Contributed as a Subject Matter Expert to **IBM Quantum Developer Certification**, authoring assessment content and validating technical accuracy for certification candidates worldwide.
- Co-designed **IBM IQC Challenge modules** with structured examples and technical documentation, improving participant comprehension and engagement with advanced quantum computing topics.

PROJECTS

Fellowship: Delta Analytics, USA *Feb 2026 to Jun 2026*
Data Fellow – Tazama Project

- Analyzed Tazama's fraud alert data to close a gap where flagged transactions carried no visibility into which of the platform's 31 rules triggered them.
- Developed a Jupyter notebook that identifies the specific rule and transaction context behind each alert, giving investigators a direct path from alert to root cause

Fellowship: **Delta Analytics,** USA
Data Fellow - Project Sanctuary Project *Feb 2025 to Jun 2025*

- Analysing **Project Sanctuary's longitudinal survey data,** working to provide a robust understanding of their impact on veteran families
- Developing dashboards which show surveys from the retreats, demographics in the USA where the retreats have happened over the years, by connecting **Salesforce database** to **Power BI**

Fellowship: **Delta Analytics,** USA
Data Fellow - TeenSmart International Project *Feb 2023 to Jul 2023*

- Designed predictive models to assess high-risk behaviours (Alcohol usage, Early Sex, Obesity, and Fighting) among Latin American teens.
- Implemented **Machine Learning Life Cycle (MLLC)** on **AWS,** using **DataBrew** for data extraction, **PySpark** for cleaning, and **SageMaker** for ML experimentation & deployment.
- Built an **AWS QuickSight Dashboard** (business intelligence) to recommend clinical interventions based on predicted risk scores, while visualising model performance metrics.

EDUCATION

University of Maryland Baltimore County	**Majors**: Data Science | GPA: 4.0	**Graduation Date:** 2021
Chaitanya Bharathi Institute of Technology	**Majors:** Information Technology | GPA: 8.2	**Graduation Date:** 2019